| | | |
|---|---|---|
| 1(a) | $t = \sqrt{(2s / g)}$<br><br>$= \sqrt{[(2 \times 36) / 9.81]}$ | **C1** |
| | $= 2.7$ s | **A1** |
| 1(b) | • reaction time between hearing the splash and stopping the stop-watch<br><br>• the sound (of the splash) takes time to reach the student **or** the stone hits the water at a different time to the sound being heard **or** the sound (of the splash) has to travel to the student<br><br>• the student might not let go of the stone from ground level<br><br>• the student might not let go of the stone and start the stop-watch at the same time<br><br>• stop-watch may not be properly calibrated / has a zero error<br><br>• (local value of) $g$ is not (exactly) 9.81 (m s$^{-2}$)<br><br>• stone given initial velocity / initial velocity not zero<br><br>• stone does not fall (exactly) vertically / in a straight line<br><br>*Any three points, 1 mark each* | **B3** |
| 1(c) | precise: results are close together / have little scatter | **B1** |
| | not accurate: the values are not close to / 50% different / (very) different from the true value | **B1** |